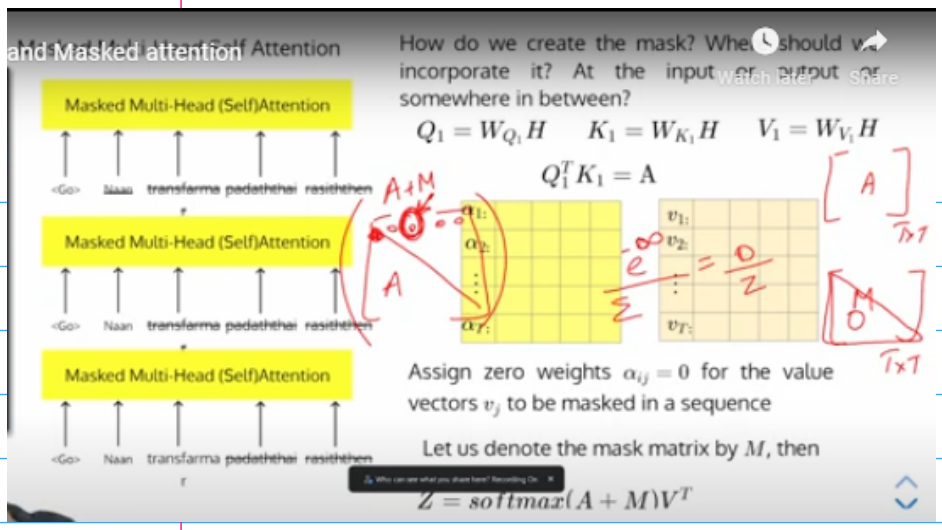
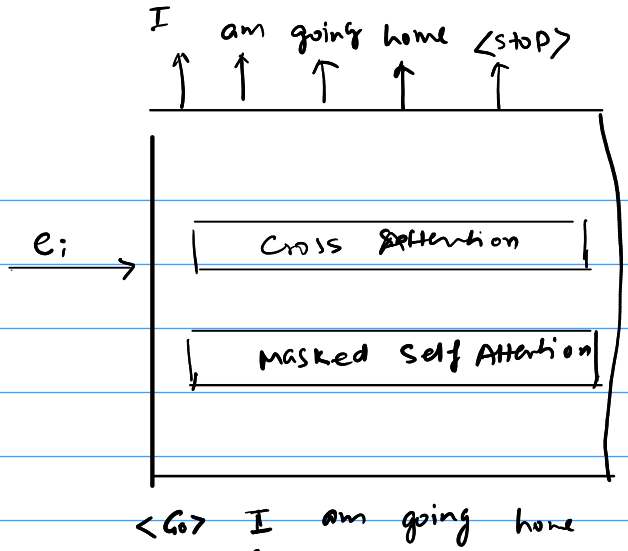




Teacher Enforcing → For some 'epochs' feed correct inputs (during training only)

W_Q, W_K, W_V for masked self attention

Lets say input dimension is 4 with embedding dimension 64

$Q = h \cdot W_Q = 5 \times 32 \times 32 \times 32 = 5 \times 32$ K and V also of same dim.

$A = Q \cdot K^T = 5 \times 32 \times 32 \times 5 = 5 \times 5$

M = 5×5 matrix with 0 and $-\infty$

$$M = \begin{bmatrix} 0 & -\infty & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty & -\infty \\ 0 & 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{5\times5}$$

Score = Softmax $\left(\frac{A+M}{\sqrt{dk}}\right)$ = 5×5 size

$S = Score \cdot V$

$5 \times 5 \times 5 \times 32 = 5 \times 32$

<Go> — — — — — — — — 32

I — — — — — — — — 32

am — — — — — — — — 32

going — — — — — — — — 32

home — — — — — — — — 32

$$S = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \end{bmatrix} \quad \text{Os} \quad 5 \times 32$$

s_1 ↓ has semantic relation of only <Go>

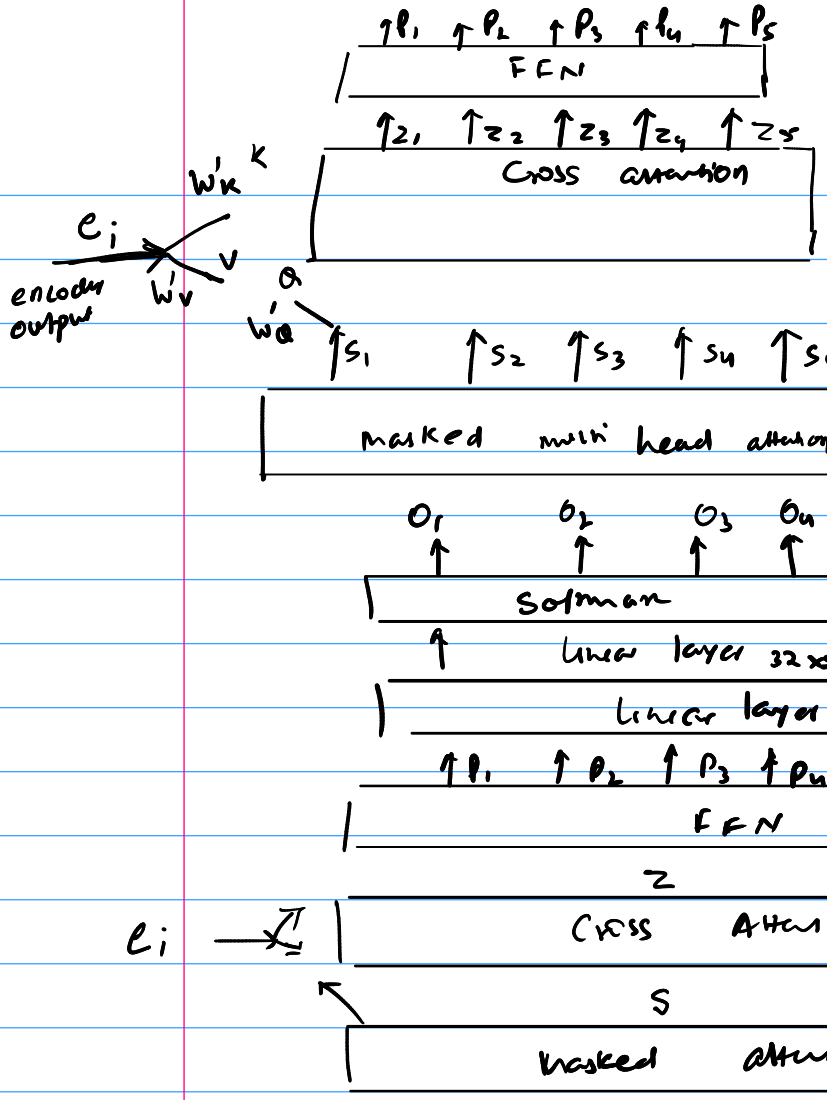
s_2 ↓ has semantic relation of I and <Go> only

s_3 — — — — — —

s_4

s_5

This S passes through Projection matrix W_0 (32×32) to get 5 × 32



$Q = S \cdot W'_Q$

$K = e \cdot W'_K$

$V = e \cdot W'_V$.

$\text{Score} = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_K}}\right)$

$Z = \text{Score} \cdot V$. $\quad 5 \times 32$

Linear layer size = 32 × Vocab. size (of decoder)

eg. 512.

~~Masked~~ attention also happens in Encoder.

eg. Context_length is 64

but input is of size 30 only.

Then 34 places are filled with $-\infty$.

Positional Encoding : In Encoder, order of words in input does not matter.

I am going home.

I going am home am.

So, we need "positional relationships" also amongst the words.

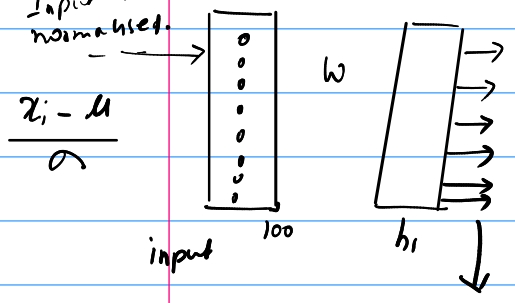
P.E$_{j,i}$ ex. I am going home and embed-dim → 4

	i = 0	1	2	3
j → input no. i → embedding ex.				
j = 0	I	—	—	—	—
1	am				
2	going				
3	home				

$$P.E_{j,i} = \begin{cases} \sin\left(\frac{j}{10000^{2i/d_{model}}}\right) & \text{if } i \text{ is even.} \\ \cos\left(\frac{j}{10000^{2i/d_{model}}}\right) & \text{if } i \text{ is odd.} \end{cases}$$

And this P.E which has same dim. as the embedding matrix, is added.

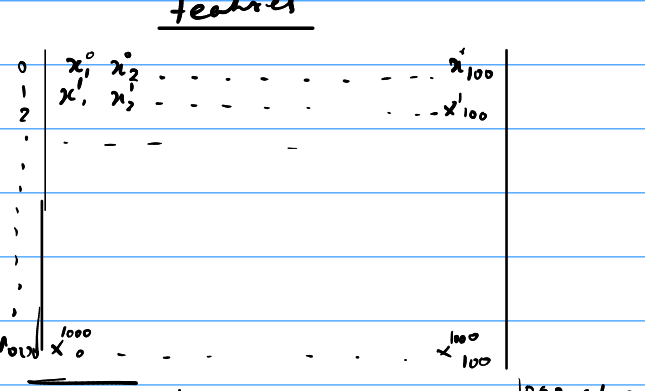
Batch Normalisation



data passed in batches.
ex. batch size - 200

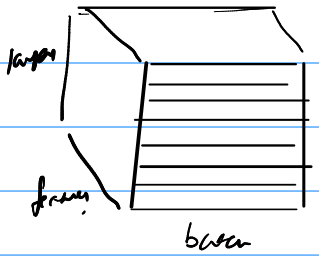
Like input Standardisation has also to as it happens in batches known as "batch Normalisation"

Training dataset



$\mu_{x_i^1} = \frac{x_i^0 + x_i^1 + \dots + x_{100}^{1000}}{1000}$, $\sigma^2 = \frac{1}{1000}\sum(x_i^1 - x^{1\mu})$

$x_i^1 = \frac{x_i^1 - \mu}{\sigma}$



In transformers

We pass let's say "I am going home" 4 × 32 input

and get output also 4 × 32. (from one sub layer — MHA / FFN)

So here also we do normalisation, but we do layer normalisation.

Plus we add residual too and then do layer normalisation.

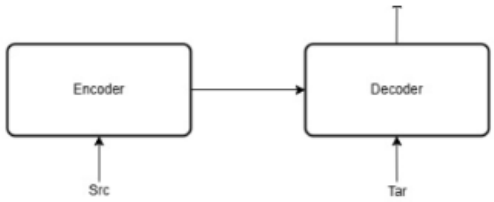
Along with $x_i' = \frac{x_i - \mu}{\sigma}$

we also do. $x_i'' = \gamma x_i' + \beta$ — bias.

dimension of γ & β is embed_dim.

∴ extra Parameters at each sub layer.

GAs



Source vocabulary size, $|V_s|$: 1000 ✓
Target vocabulary size, $|V_t|$: 1500 ✓
The context length, T: 128 ✓
The embedding dimension, $d_{embed} = d_{model}$: 64 ✓
The projection dimensions, $d_q = d_k = d_v$: 16 ✓
The number of heads, n_h: 4
The hidden dimension of FFNN, d_{ff}: 256
Uses layer norm and residual connection

We pass the sample from the source language ``I enjoyed the movie Transformer" to the encoder and its translation ``Naan transformer padaththai rasithen" to the decoder. All the learnable parameters are randomly initialized. The model uses the teacher-forcing approach throughout the training process.

1) Assume the model has two layers ($N = 2$). Calculate the total number of parameters in the model (excluding the embedding layer and output layer). Moreover, no bias was added to the neuron in the FFNN layers.

Encoder. One layer. W_Q, W_K, W_V, W_O (MHA)
normalisation Parameters (MHA)
W_1, W_2 (FFN)
normalisation Parameters (FFN)

Encoder MHA. W_Q, W_K, W_V, W_O → Size $\frac{64 \times 64 \times 4}{64+64}$ = 16384 = 128

Normalisation. →

FFN. $64 \times 256 + 256 \times 64$ = 32768

Normalisation → $64 + 64$ = 128

∴ Total no. of Parameters in Encoder in One layer : 49408
=
=

∴ Total $E_p = 2 \times 49408 =$ 98816 ✓

Now Decoder.

Multi head attention — W_Q, W_K, W_V, W_O — $4 \times 64 \times 64 = 16384$

normalisation layer. — $64 + 64 = 128$

Cross Attention — $W_Q, W_K, W_V, W_O \rightarrow 4 \times 64 \times 64 = 16384$

normalisation layer. $\rightarrow 64 + 64 = 128$

FFN $\rightarrow$ W_1, , W_2 $\rightarrow 64 \times 256 \times 2 = 32768$

normalisation layer. $\rightarrow 64 + 64 = 128$

∴ Total no of 'p' in one decoder layer = 65920

=

=

∴ in 2 layers = 65920×2 = 131840

∴ Total no. of p in model = 98816 + 131840

= 230656 ✓

1) Assume the model has two layers ($N = 2$). Calculate the total number of parameters in the model (excluding the embedding layer and output layer). Moreover, no bias was added to the neuron in the FFNN layers.

230656.

1

2) How many parameters does the output layer have?

64×1500 ($d_{model} \times Vocab_size(decoder)$

= 96000

1

3) How many parameters do the embedding layers have?

embedding layer $\rightarrow$ 1000×64 = 64000

4) At time step $t = 1$, the prediction probability for the words ``Nann", ``transformer" and ``padaththai" are 0.55,0.15 and 0.2, respectively. What is the probability for the word ``rasithen"? If you think, the given information is insufficient enter -1

= $1 - (0.55 + 0.15 + 0.2)$

= $1 - 0.9 = 0.1$

Decoder output is probability distribution.

5) The question is the continuation of the last question from the previous week].

(I used numpy for this)

The input embeddings for the words ``learning", ``brings" and ``joy" are $h_1 = [0.5, 0.25, 1], h_2 = [0.1, 0.25, 0]$, and $h_3 = [0.1, 0.1, 0.9]$, respectively. Note that the embeddings are row vectors. The projection matrices are as follows $W_Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$ $W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}$

The following quantities are computed as >$Q = HW_Q$ $K = HW_K$ $V = HW_V$ Let e_j denote the unnormalized attention score, a_j denote the normalized attention score (ignore the scaling by $\sqrt{d_k}$) and z_j denote the linear combination of the value vectors for the $j-th$ word. Suppose the gradient vector $\frac{\partial L}{\partial z_1} = [1, 1]$, then what is the gradient vector $\frac{\partial L}{\partial e_1}$? Enter the sum of gradients.

$\frac{\partial L}{\partial z_1} = [1,1]$ $\frac{\partial L}{\partial e_1}$?

Note: Keep only two significant digits after the decimal point in all your calculation

learning brings joy

learning	0.5	0.25	1
brings	0.1	0.25	0
joy	0.1	0.1	0.9

3x3.

$W_Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$ $W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}$

$Q = h W_Q$ (3x3 · 3x2) $= \begin{vmatrix} 0.5 & 0.25 & 1 \\ 0.1 & 0.25 & 0 \\ 0.1 & 0.1 & 0.9 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 0.25 & 1.75 \\ -0.15 & 0.35 \\ 0 & 1.1 \end{vmatrix}$ (3x3 · 3x2)

$K = \begin{vmatrix} 0.5 & 0.25 & 1 \\ 0.1 & 0.25 & 0 \\ 0.1 & 0.1 & 0.9 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 0.25 & 1.5 \\ 0.25 & 0.1 \\ 0.1 & 1.0 \end{vmatrix}$

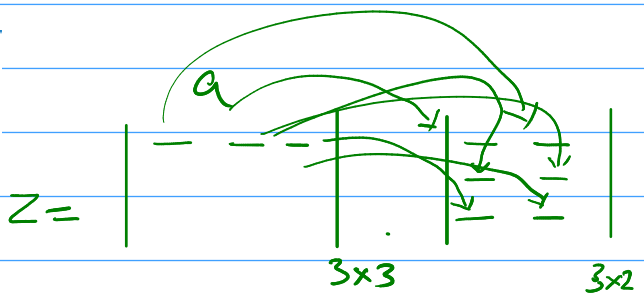
$V = \begin{vmatrix} 0.5 & 0.25 & 1 \\ 0.1 & 0.25 & 0 \\ 0.1 & 0.1 & 0.9 \end{vmatrix} \begin{vmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{vmatrix} = \begin{vmatrix} 0.75 & 0.75 \\ -0.25 & -0.25 \\ 0.8 & 0.8 \end{vmatrix}$

$e_j = Q \cdot K^T =$

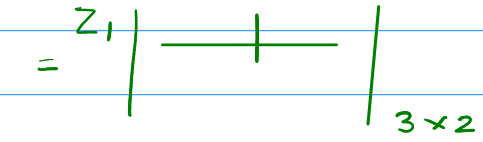
```
[[ 2.69  0.24  1.77]
 [ 0.49 -0.    0.33]
 [ 1.65  0.11  1.1 ]]
```

a = normalised e_j = Softmax (e_j) without using $\sqrt{d_k}$.

a =
```
[[0.67 0.06 0.27]
 [0.4  0.25 0.35]
 [0.56 0.12 0.32]]
```



now $Z = a \cdot V$ =
```
[[0.71 0.71]
 [0.52 0.52]
 [0.65 0.65]]
```

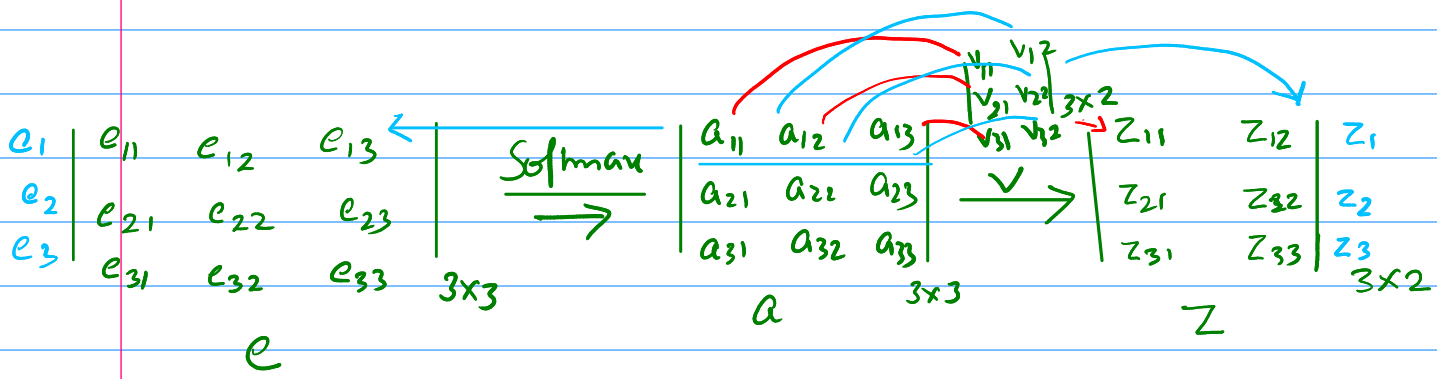


Given $\frac{\partial L}{\partial Z_1} = [1, 1]$. , find $\frac{\partial L}{\partial e_1}$

$\frac{\partial L}{\partial e} = \frac{\partial L}{\partial Z} \cdot \frac{\partial Z}{\partial a} \cdot \frac{\partial a}{\partial e}$ Size 3 x 3.

$Z = a \cdot V \quad \therefore \frac{\partial Z}{\partial a} = V$

$a = \text{Softmax}(e) \quad \therefore \frac{\partial a}{\partial e} = a(1-a) = a(1-a)$



$\frac{\partial L}{\partial e_{1\ 1\times 3}} = \frac{\partial L}{\partial Z_1}_{1\times 2} \cdot V_{3\times 2}$ $= \frac{\partial L}{\partial Z_1} \cdot V^T \rightarrow 1\times 3$

$\left(\frac{\partial L}{\partial Z_1} \cdot V^T\right)_{1\times 3} = \underline{1\times 3}$

$$\frac{\partial L}{\partial e} = \underbrace{\left(\frac{\partial L}{\partial z_1} \cdot V^T\right)}_{1\times 3} \underbrace{a(1-a)}_{3\times 3} = \underbrace{\quad}_{1\times 3}$$

$$= (1,\ 1)_{1\times 2} \begin{vmatrix} 0.75 & 0.75 \\ -0.25 & -0.25 \\ 0.8 & 0.8 \end{vmatrix}^T$$

```
[[0.67 0.06 0.27]
 [0.4  0.25 0.35]
 [0.56 0.12 0.32]]
```

$$= \quad /$$

$$x = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}$$

$$y = \text{Softmax}(x) = \begin{pmatrix} \frac{e^{x_1}}{e^{x_1}+e^{x_2}} & \frac{e^{x_2}}{e^{x_1}+e^{x_2}} \\ \frac{e^{x_3}}{e^{x_3}+e^{x_4}} & \frac{e^{x_4}}{e^{x_3}+e^{x_4}} \end{pmatrix} = \begin{pmatrix} y_1 & y_2 \\ y_3 & y_4 \end{pmatrix}$$

$\frac{\partial y}{\partial x} = y_i(1-y_i)$ for diagonal elements

and $-y_i y_j$ for non diagonal elements.

```
a = [[0.67 0.06 0.27]
     [0.4  0.25 0.35]
     [0.56 0.12 0.32]]
```

```
e = [[ 2.69  0.24  1.77]
     [ 0.49 -0.    0.33]
     [ 1.65  0.11  1.1 ]]
```

$a_1 = [0.67 \quad 0.06 \quad 0.27]$, $e_1 = [2.69 \quad 0.24 \quad 1.77]$

$\frac{\partial a_1}{\partial e_1}$ will be a Jacobian matrix

if $a = [s_1 \quad s_2 \quad s_3]$, then $\frac{\partial a}{\partial e} = \begin{bmatrix} s_1(1-s_1) & -s_1 s_2 & -s_1 s_3 \\ -s_2 s_1 & s_2(1-s_2) & -s_2 s_3 \\ -s_3 s_1 & s_3 s_2 & -s_3(1-s_3) \end{bmatrix}$

3×3

$S_1 \quad S_2 \quad S_3$

$$R = \begin{bmatrix} 0.67 & 0.06 & 0.27 \\ 0.4 & 0.25 & 0.35 \\ 0.56 & 0.12 & 0.32 \end{bmatrix}$$

$$\therefore \frac{\partial a_1}{\partial e_1} = \begin{vmatrix} 0.67\times0.33 & -0.67\times0.06 & -0.67\times0.27 \\ -0.06\times0.67 & 0.06(0.94) & -0.06\times0.27 \\ -0.27\times0.67 & -0.27\times0.06 & 0.27(1-0.27) \end{vmatrix}$$

$$= \begin{bmatrix} 0.22 & -0.04 & -0.18 \\ -0.04 & .05 & -0.02 \\ -0.18 & -0.02 & 0.2 \end{bmatrix}$$

$$\therefore \frac{\partial L}{\partial e_1} = \left[(1,1)_{1\times2} \begin{bmatrix} 0.75 & 0.75 \\ -0.25 & -0.25 \\ 0.8 & 0.8 \end{bmatrix}^T \right] \cdot \begin{bmatrix} 0.22 & -0.04 & -0.18 \\ -0.04 & .05 & -0.02 \\ -0.18 & -0.02 & 0.2 \end{bmatrix}$$

$$= |1,1|_{1\times2} \begin{vmatrix} 0.75 & -0.25 & 0.8 \\ 0.75 & -0.25 & 0.8 \end{vmatrix}_{2\times3} \cdot \begin{bmatrix} 0.22 & -0.04 & -0.18 \\ -0.04 & .05 & -0.02 \\ -0.18 & -0.02 & 0.2 \end{bmatrix}$$

$$= \begin{vmatrix} 1.5 & -0.5 & 1.6 \end{vmatrix}_{1\times3} \begin{vmatrix} 0.22 & -0.04 & -0.18 \\ -0.04 & 0.05 & -0.02 \\ -0.18 & -0.02 & 0.2 \end{vmatrix}$$

3

$$\frac{\partial L}{\partial e_1} = \left[(0.33+0.02-0.28), (-0.06-0.025-0.032), (-0.27+0.01+0.32) \right]$$

$$0.350 - 0.33 = 0.02$$
$$0.288 - 0.27 = 0.018 \approx 0.02$$
$$0.062 - 0.06 = 0.002$$

$$= \left[0.062, -0.127, 0.06 \right]$$

$$\therefore \text{Sum} = 0.062 + 0.06 - 0.127$$
$$= 0.122 - 0.127 = -0.007$$
$$\approx 0$$

6) Which of the following is (are) the hyper-parameters?

- [x] vocabulary size
- [x] (d_k, d_v, d_q)
- [x] number of heads
- [x] warm-up step
- [x] d_{model}
- [x] number of layers
- [ ] size of the attention mask

hyperparameters — Parameters of the model.

Vocab-size ✓

d_k, d_v, d_q ✓

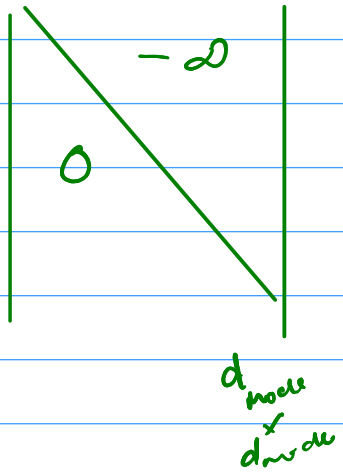
no. of heads ✓

warm up step ✓

d_{model} ✓

no. of layers ✓

size of the attention mask ✗ → It is automatically done: M =



d_{model} ✓ d_{model}

7) The table below shows the input samples and the corresponding translation produced by the model.

Source	Target
I enjoyed the movie Transformers	நான் டிரான்ஸ்பார்மர்ஸ் திரைப்படத்தை ரசித்தேன்
We live among its people now	நாம் இப்பொழுது அவர்களிடையே வாழ்கிறோம்
They have much to learn	அவர்கள் கற்றுக்கொள்ள நிறைய இருக்கிறது
Freedom is the right of all sentient beings	சுதந்திரம் என்பது அனைத்து உணர்வுள்ள உயிரினங்களின் உரிமை

To produce the complete translation for the third sample in the table, we would have to run the decoder of the model at least ____ times.

encoder will give e in one go.

for decoder to decode third sample

They have much to Learn

and target has 4 words.

So, to Complete full translation of the sentence, decoder has to run at least 4 times.